



Fake it 'till you make it

**Zarr-like Access of Existing NetCDF4
Datasets**

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The Problem

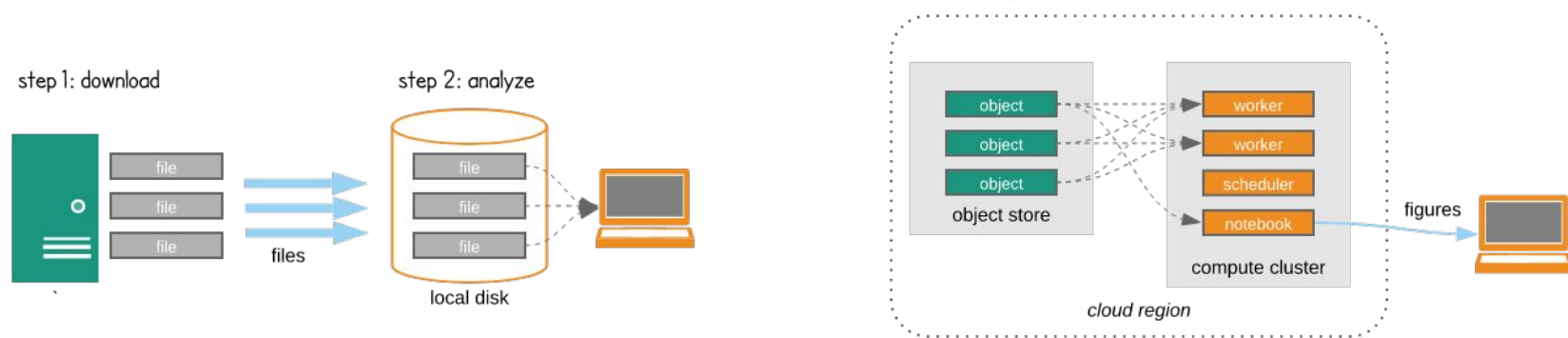
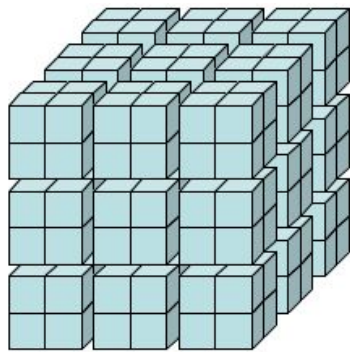


Image Source - Pangeo/Ryan Abernathy

- Peta(Exa?)bytes of geophysical data now available in the cloud
- Much of the data are in NetCDF4/HDF5 format
 - Not efficient for cloud computing

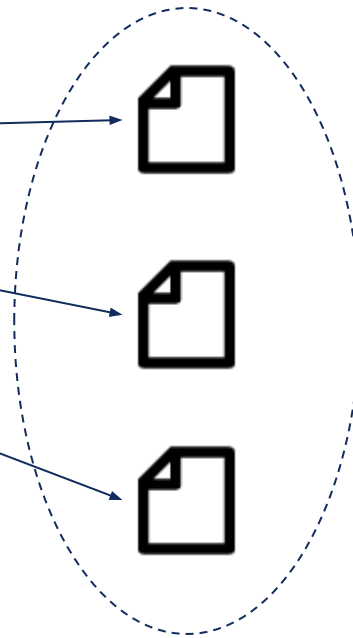
NetCDF vs Zarr

NetCDF



Single File Object

Zarr



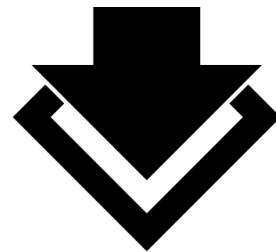
Multiple File Objects
(One per chunk)



Plaintext
Metadata
File

ReferenceMaker/ReferenceFileSystem:

- Part of Intake group's `fsspec` project
- Access existing NetCDF/HDF files *as if they are Zarr*
- Zarr-like Metadata files (few MB each)
 - Remote file address (Azure Blob, AWS S3, etc)
 - Data variable and chunking info
 - Can be read as a Zarr file
- Extracts byte-ranges for individual variables/chunks
 - Instead of individual chunk files in Zarr

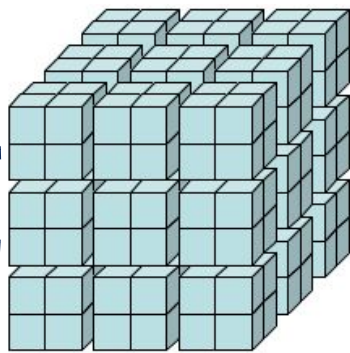


INTAKE

ReferenceMaker

NetCDF

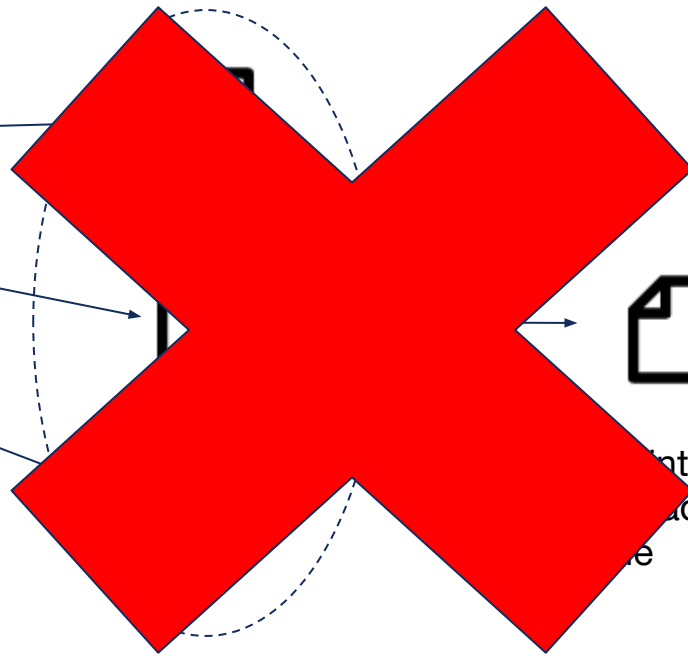
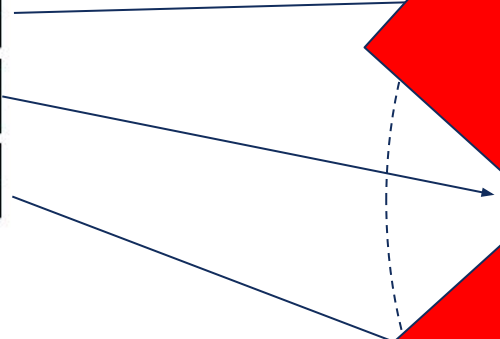
Zarr



Single File Object



ReferenceMaker
Metadata



Multiple File Objects
(One per chunk)



Context
data
e

Creating JSONS

`@dask.delayed` ←

Easily distributed with Dask!

```
def gen_json(url):  
    so = dict(  
        mode="rb", anon=True, default_fill_cache=False, default_cache_type="none"  
    )
```

```
    outpath = './jsons/' + url.split('/')[-1].split('.nc')[0] + '.json'
```

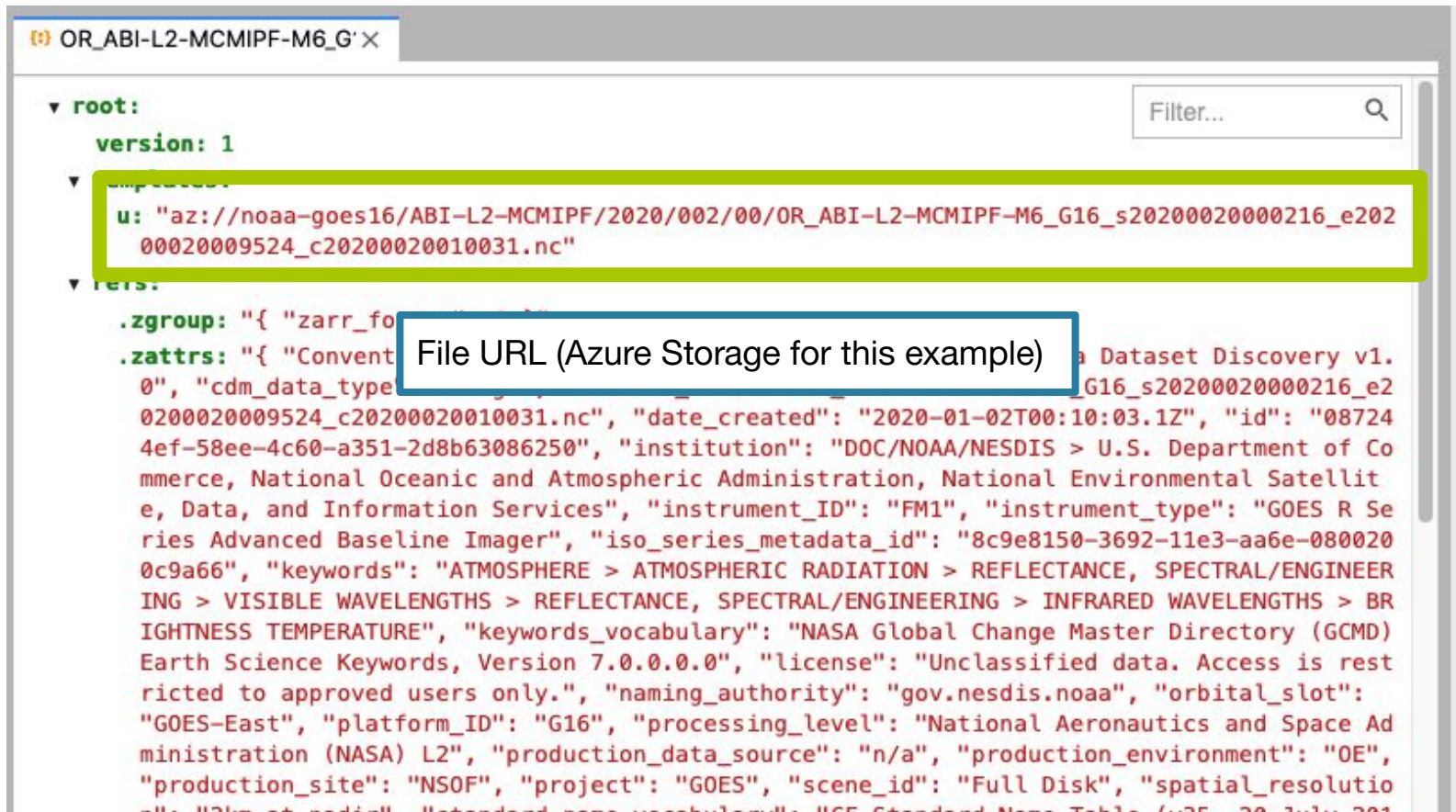
```
    with fsspec.open(url, **so, account_name='goesuwest') as inf:  
        h5chunks = SingleHdf5ToZarr(inf, u, inline_threshold=300)
```

Read remote file and
process chunks

```
    with open("./reference.json", 'wb') as outf:  
        outf.write(ujson.dumps(h5chunks.translate()).encode())
```

Write metadata to file

Reference JSON Example



The screenshot shows a JSON viewer interface with a tab labeled "OR_ABI-L2-MCMIPF-M6_G". The JSON structure is as follows:

```
root:
  version: 1
  u: "az://noaa-goes16/ABI-L2-MCMIPF/2020/002/00/OR_ABI-L2-MCMIPF-M6_G16_s20200020000216_e20200020009524_c20200020010031.nc"
  .zgroup: "{ \"zarr_format\": \"2.0\" }"
  .zattrs: "{ \"Conventions\": \"GOES-R Standard Name Table (v25-00-01)\" }
```

A green rectangular highlight is placed around the value of the "u" field. A blue-bordered callout box with the text "File URL (Azure Storage for this example)" points to this highlighted value.

Reference JSON Example

```
ministration (NASA) L2 , production_data_source : n/a , production_environment : OE ,  
"production_site": "NSOF", "project": "GOES", "scene_id": "Full Disk", "spatial_resolution":  
n": "2km at nadir", "status": "Final", "table_name": "Table (v35, 20 July 2016)", "summary": "Multi-  
ducts are digital maps of the Earth's surface and Moisture Imagery Products at visible, near-IR,  
and IR bands.", "time_coverage_start": "2020-01-02T00:00:21.6Z", "time_coverage_end": "2020-01-02T00:00:21.6Z", "timeline_id": "ABI Mode 6", "title": "ABI L2 Cloud and Moisture Imagery" }
```

Variable (CMI_C01) chunk info,
compression, shape, etc

```
CMI_C01/.zarray: "{ \"chunks\": [ 226, 226 ], \"compressor\": { \"id\": \"zlib\", \"level\": 1 }, \"dtype\": \"<i2\", \"fill_value\": -1, \"filters\": null, \"order\": \"C\", \"shape\": [ 5424, 5424 ], \"zarr format\": 2 }"
```

```
CMI_C01/.zattrs: "{ \"_ARRAY_DIMENSIONS\": [ \"y\", \"x\" ], \"_FillValue\": -1, \"_Netcdf4Dimid\": 0, \"_Unsigned\": \"true\", \"add_offset\": 0.0, \"ancillary_variables\": \"DQF_C01\", \"cell_methods\": \"t: point area: sum (interval: 0.000028 rad)\", \"coordinates\": \"band_id_C01 band_wavelength_C01 t y x\", \"downsampling_method\": \"average\", \"grid_mapping\": \"goes_imager_projection\", \"long_name\": \"ABI Cloud and Moisture Imagery reflectance factor\", \"resolution\": \"y: 0.000056 rad x: 0.000056 rad\", \"scale_factor\": 0.00031746001332066953, \"sensor_band_bit_depth\": 10, \"standard_name\": \"toa_lambertian_equivalent_albedo_multiplied_by_cosine_solar zenith_angle\", \"units\": \"1\", \"valid_range\": [ 0, 4095 ] }"
```

```
▼ CMI_C01/0.4: [] 3 items  
0: "{u}"  
1: 253643  
2: 468
```

Single Chunk Byte Range

Reference JSON Example

Set “reference” filesystem
and specify JSON file

```
fs = fsspec.filesystem('reference',  
                        fo='reference.json',  
                        remote_protocol='az',  
                        remote_options={'account_name': 'goeseuwest'},  
                        simple_templates=True)
```

Specify remote file
protocol (Azure here) and
authentication

```
m = fs.get_mapper("")  
ds = xr.open_dataset(m, engine='zarr')
```

Pass mapper to xarray and
specify Zarr engine

GOES Workflow



Data stored in same region as compute



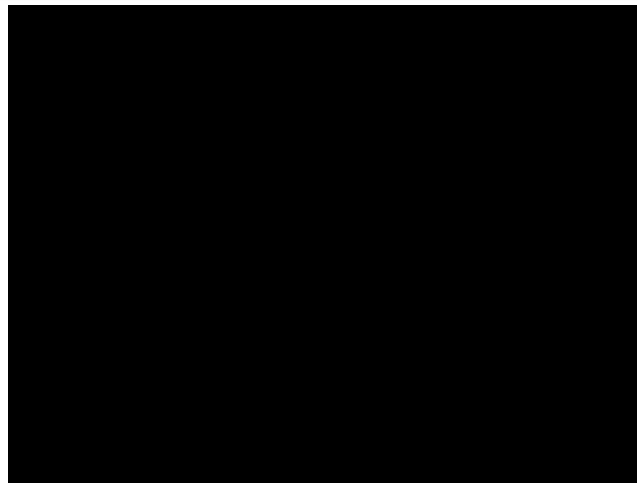
4 CPU, 32 GB Memory,
8 Dask Workers

- Created a test workflow on Microsoft Planetary Computer
- Analysis of 24 hours of GOES data
- Test access to data stored in Azure in 3 formats:
 - NetCDF4
 - NetCDF4 + ReferenceMaker
 - Zarr

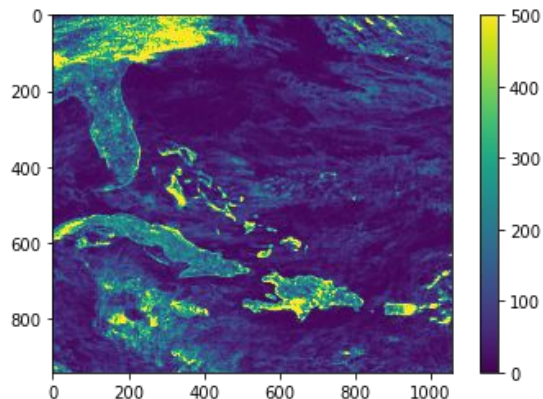
GOES Workflow



Data stored in same region as compute



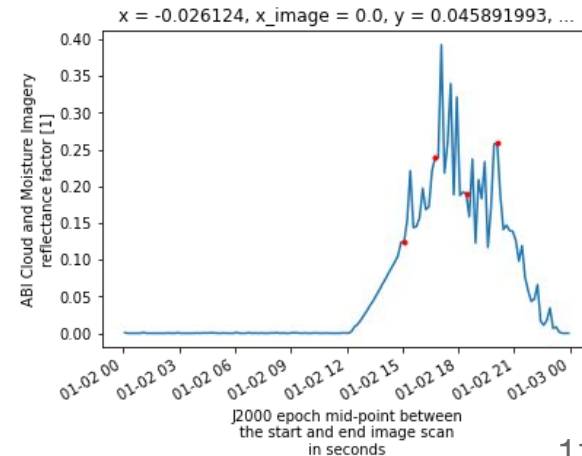
4 CPU, 32 GB Memory,
8 Disk Workers



True Color 2020-01-02 0005 UTC



Veggie Color 2020-01-02 0005 UTC



GOES Processing Times

For 24 hours of GOES-16 “ABI-L2-MCMIPF” product data

<u>Format</u>	<u>Preprocess Time</u>	<u>Data Open Time</u>	<u>Workflow Time</u>	<u>Extra Storage</u>
Remote NetCDF4	0 min	10 minutes	40 min	0 GB
Zarr	1 h 38 min	30 seconds	4 min	52 GB
ReferenceFileSystem	1 h 25 min	35 seconds	5 min 30 s	<u>416 MB</u>

- NetCDF has no upfront preprocess time cost
 - But very slow open and workflow times
- Zarr comes at cost of high storage costs (full data duplication)
- ReferenceFileSystem has speed of Zarr but **≤1% storage cost**

This is a big deal!

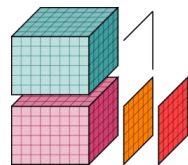
- ReferenceMaker/FileSystem harnesses the cloud-optimization of Zarr without converting any data
- Allows current institutions to cheaply make existing cloud-hosted data more optimized
- Reference files can be also created, hosted, and shared by third parties
- Automatable
 - e.g. Pangeo Forge

If you are a scientist that works with these tools

- Join the Pangeo community
- Talk to project maintainers about your workflow needs
- Open issues/bug reports on GitHub
- Submit pull requests, even if your code isn't 100% ready

PANGEO

Microsoft Planetary
Computer



xarray



Iris



Acknowledgements

Thank you to the ReferenceFileSystem team!

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- <https://github.com/intake/fsspec-reference-maker>

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Please feel free to reach out with questions
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Link to site with interactive code examples, tutorials, resources, and contact info

<https://lucassterzinger.com/2021-siparcs-poster/>